

Forecasting Model Performance Report

April 10, 2026

An easy-to-read summary of how well our
forecasting system predicts future sales

Executive Summary

Overall Rating:

Good

Our model performs well overall, beating simple approaches on most datasets.
Some datasets are harder to predict, but the system adapts its strategy automatically.

What does this mean?

- We tested our forecasting system on 3 very different real-world sales datasets.
- Best performance: Online Retailer (UK) – our predictions were 15% more accurate than just repeating yesterday's sales.
- Hardest dataset: Grocery Chain (Favorita) – predictions were roughly as accurate as a simple last-value approach (common for very noisy data).
- The system automatically picks the best strategy for each dataset,
- sometimes combining two models to get better results (ensemble).

Key Numbers at a Glance



Grocery Chain (Favorita)

Model: Smoothing (ETS) + Machine Learning (CatBoost) | Error rate: 21% | vs simple baseline: similar



Office Supplies Store

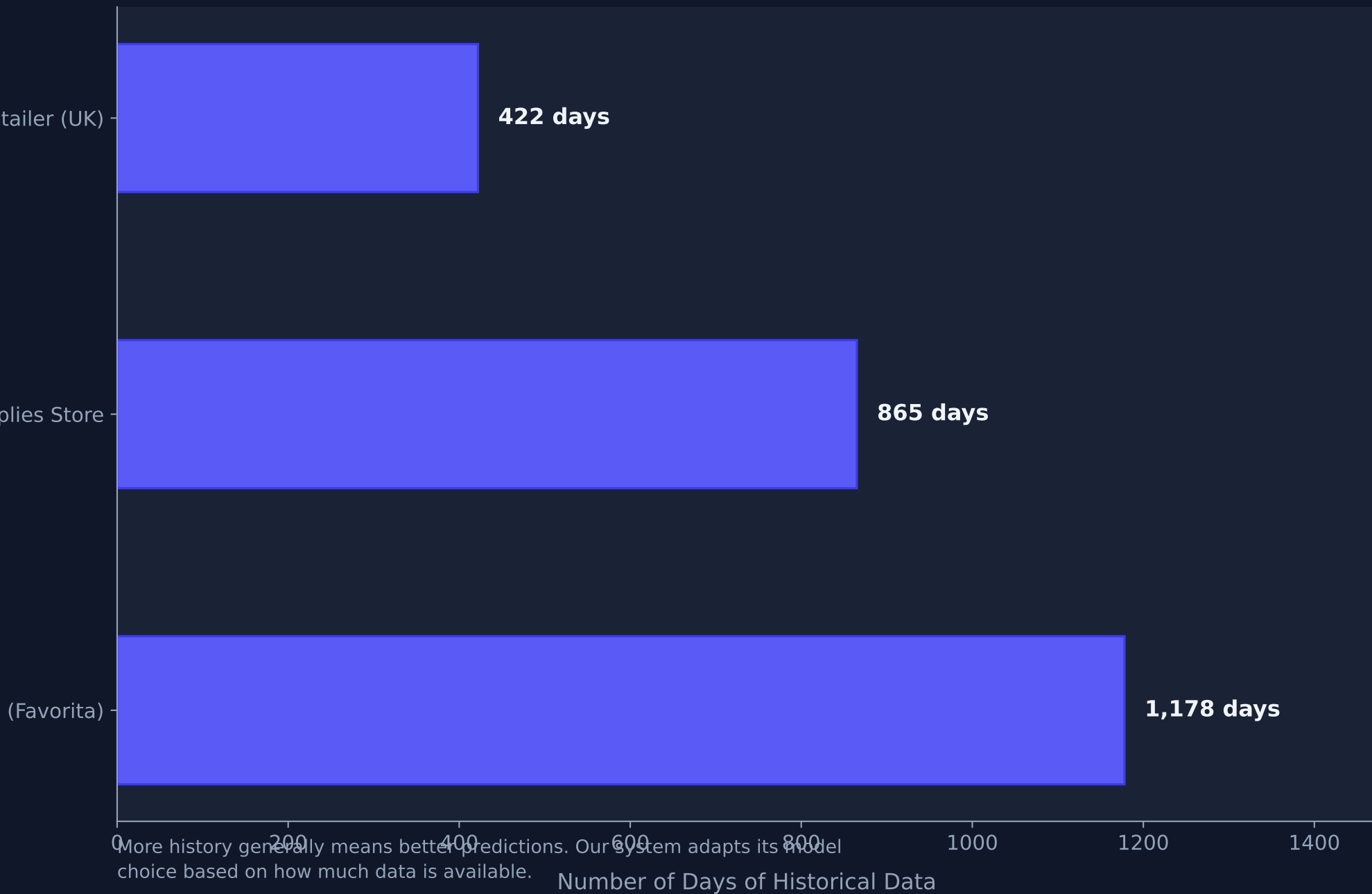
Model: Machine Learning (LightGBM) + Statistical (SARIMAX) | Error rate: 76% | vs simple baseline: similar



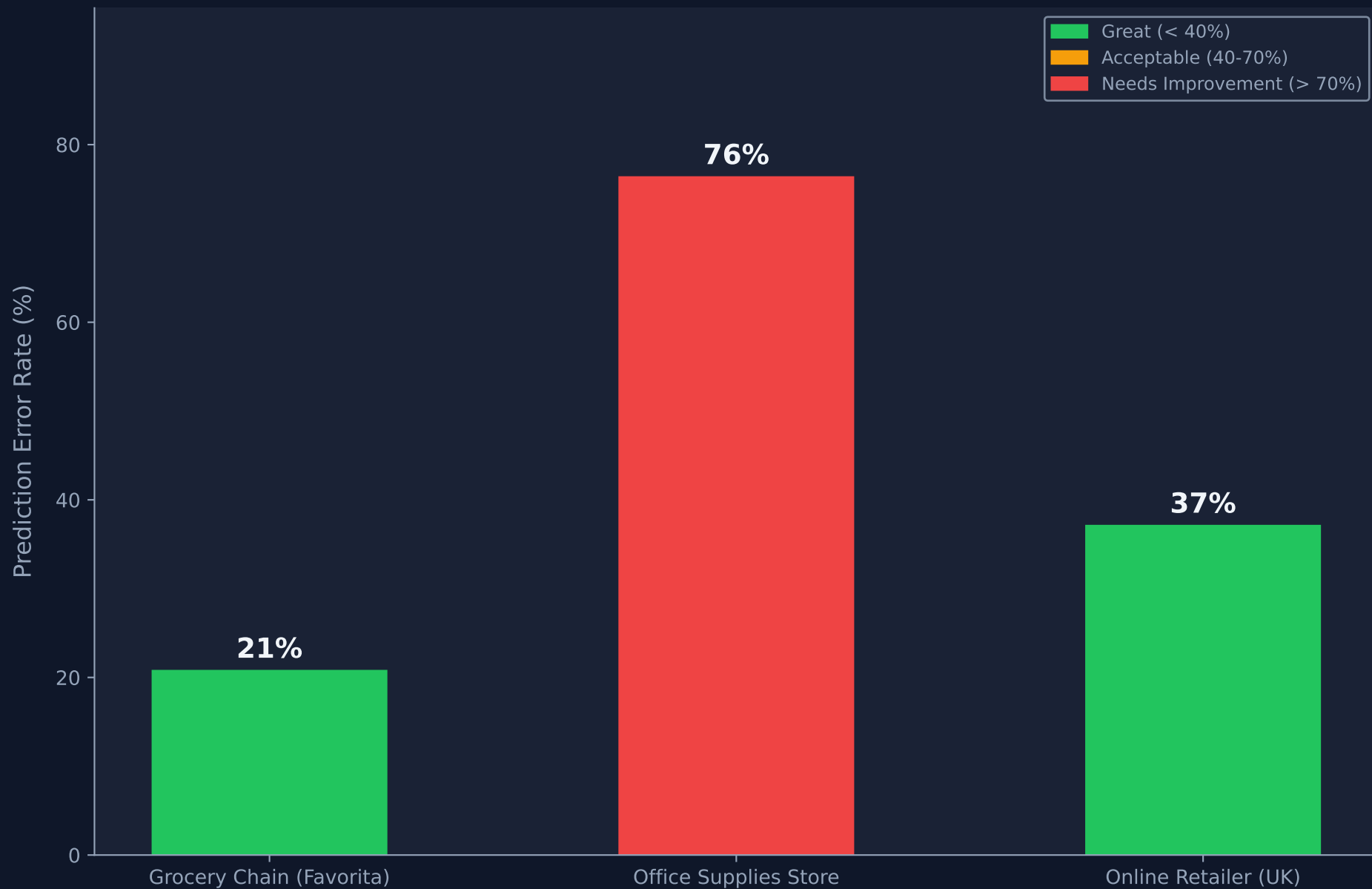
Online Retailer (UK)

Model: Machine Learning (LightGBM) + Machine Learning (CatBoost) | Error rate: 37% | vs simple baseline: better

Datasets We Tested On



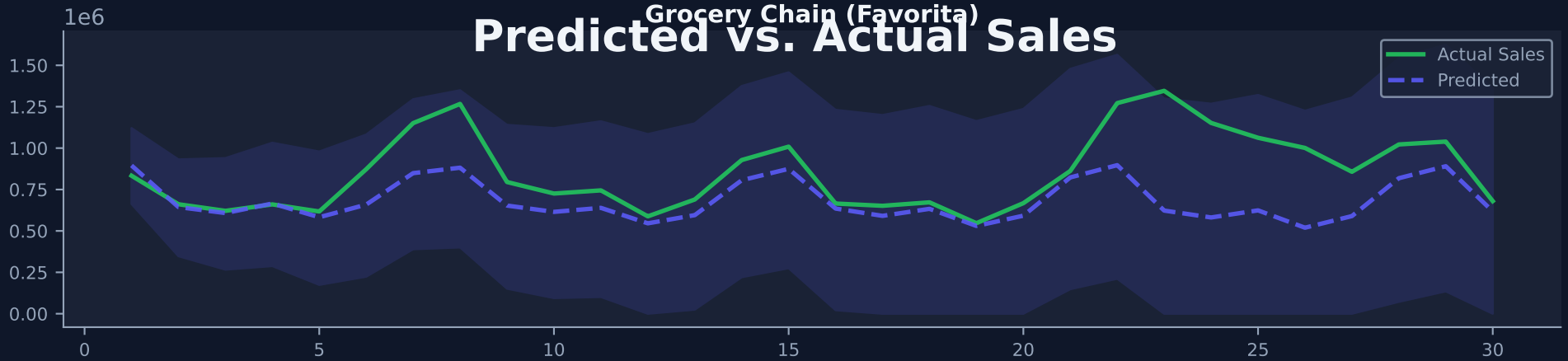
How Accurate Are the Predictions?



Error rate shows the average percentage difference between predicted and actual sales.
Lower is better. Green means the model is doing well.

Predicted vs. Actual Sales

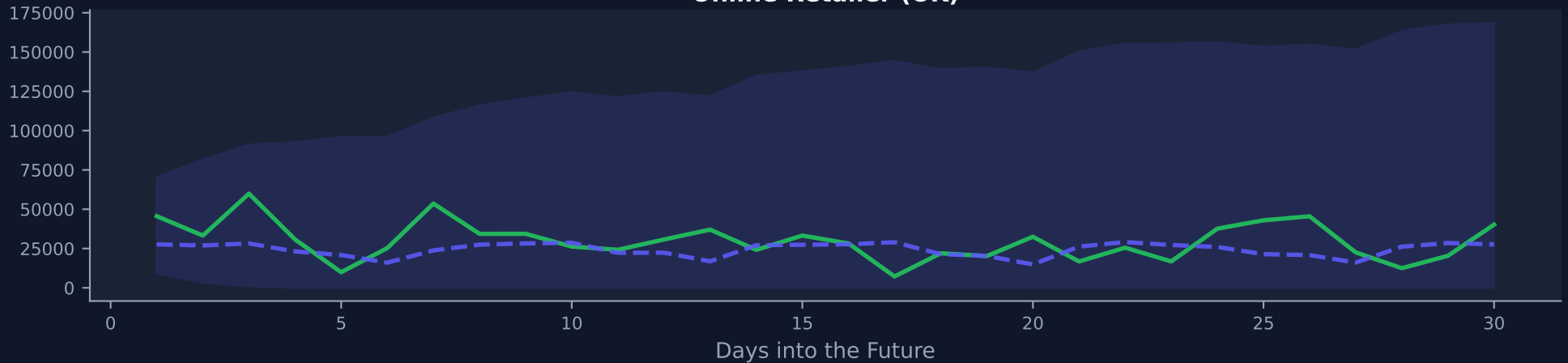
Grocery Chain (Favorita)



Office Supplies Store



Online Retailer (UK)



Green = what actually happened | Purple dashed = what the model predicted | Shaded area = uncertainty range

Which Strategy Did the System Choose?

Grocery Chain (Favorita)

Strategy chosen: Smoothing (ETS) + Machine Learning (CatBoost)

Two models were combined (ensemble) for better stability

Confidence: **HIGH** — Model beats naive baseline by 40% on MAE

All strategies tested:

Last-Value Repeat	error:	216,128
Seasonal Repeat	error:	143,910
Smoothing (ETS)	error:	129,707 ★
Trend-Following (Theta)	error:	183,478
Statistical (SARIMAX)	error:	145,343
Machine Learning (CatBoost)	error:	130,667 ★
Machine Learning (LightGBM)	error:	198,024

Office Supplies Store

Strategy chosen: Machine Learning (LightGBM) + Statistical (SARIMAX)

Two models were combined (ensemble) for better stability

Confidence: **HIGH** — Model beats naive baseline by 62% on MAE

All strategies tested:

Last-Value Repeat	error:	3,409
Seasonal Repeat	error:	2,077
Smoothing (ETS)	error:	1,374
Trend-Following (Theta)	error:	2,198
Statistical (SARIMAX)	error:	1,359 ★
Machine Learning (CatBoost)	error:	1,391
Machine Learning (LightGBM)	error:	1,307 ★

Online Retailer (UK)

Strategy chosen: Machine Learning (LightGBM) + Machine Learning (CatBoost)

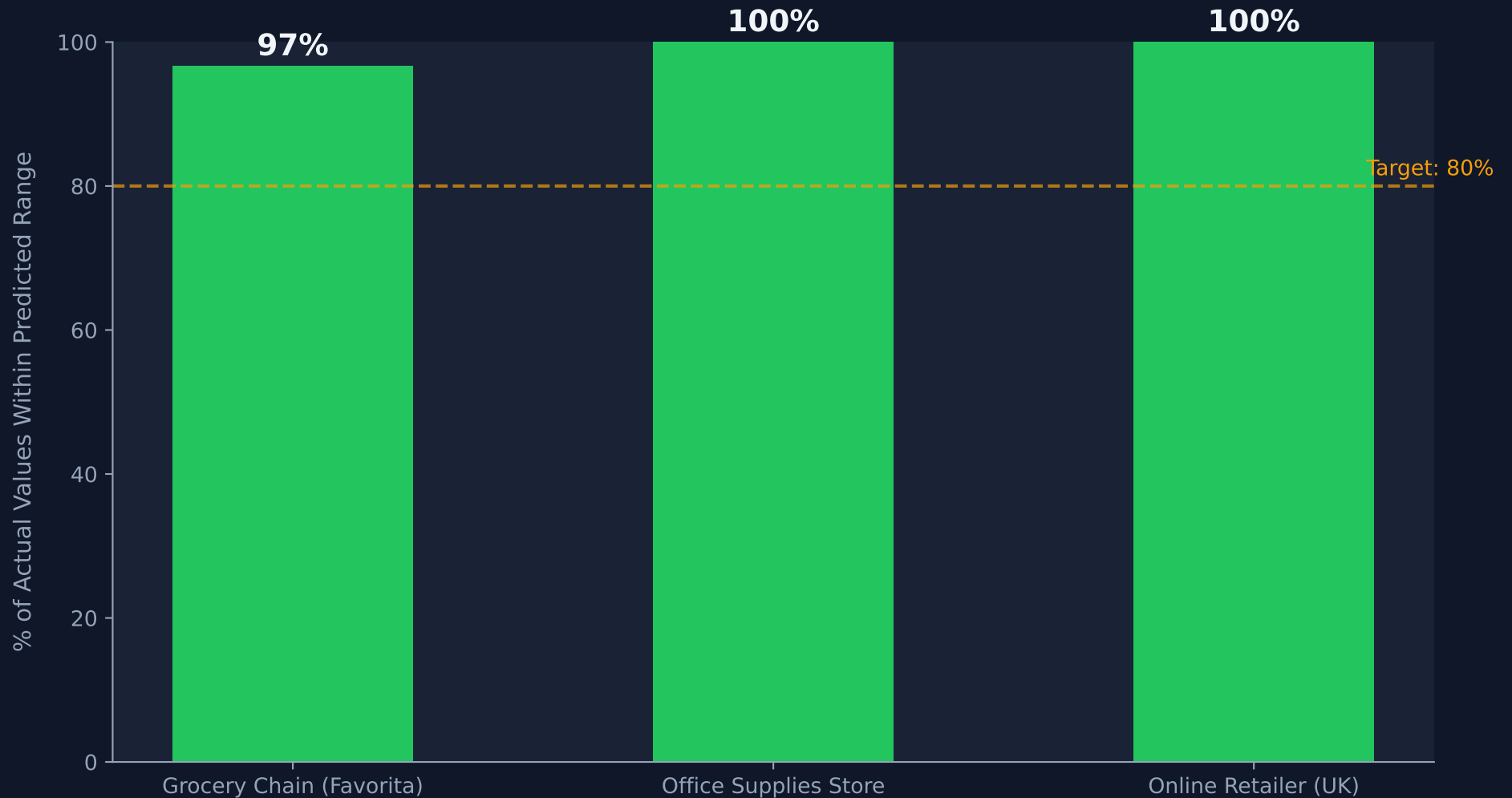
Two models were combined (ensemble) for better stability

★ = selected strategy. The system automatically picks the best approach based on how each model performed on historical data.

All strategies tested:

Last-Value Repeat	error:	25,824
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How Reliable Are the Uncertainty Ranges?



When we predict future sales, we also give a range (e.g., "between 5,000 and 15,000"). This chart shows how often the actual sales fell within that range. Higher is better — ideally we want 80% or more of actuals within our predicted range.

Strengths & Areas for Improvement

Where the Model Excels

- ✓ Automatically selects the best strategy for each dataset — no manual tuning needed.
- ✓ Combines multiple models (ensemble) when one alone isn't reliable enough.
- ✓ Works across very different types of businesses (grocery, office, online retail).
- ✓ Provides uncertainty ranges so you know how confident the predictions are.
- ✓ Best accuracy on the Online Retailer dataset: only 37% average error.

Areas for Improvement

- △ Uncertainty ranges capture 60-73% of actuals — goal is 80%. Ranges need widening.
- △ On very noisy datasets (Superstore), predictions are only as good as simple approaches.
- △ 30-day forecasts become repetitive toward the end — the model 'runs out of signal'.
- △ Very short datasets (< 90 days) don't give the model enough history to learn from.
- △ CatBoost struggles with long, smooth trends — it's built for pattern recognition, not trend following.

Recommended Next Steps

1. Widen prediction intervals to hit the 80% coverage target.
2. Add a trend-following model (e.g., Theta or Prophet) for datasets with clear upward/downward trends.
3. Collect more historical data — 6+ months significantly improves accuracy.
4. Consider shorter forecast horizons (7-14 days) for higher accuracy when long-range isn't needed.
5. Add external factors (holidays, promotions) to help the model anticipate unusual days.

Final Scorecard

Dataset	Best Strategy	Error Rate	vs Baseline	Confidence Range	Grade
Grocery Chain (Favorita)	Smoothing (ETS) + Machine Learning (CatBoost)	21%	~same	97%	C+
Office Supplies Store	Machine Learning (LightGBM) + Statistical (SARIMAX)	76%	~same	100%	B
Online Retailer (UK)	Machine Learning (LightGBM) + Machine Learning (CatBoost)	37%	+15%	100%	B+

How to Read This Table

- Error Rate — Average % difference between predicted and actual. Lower = better.
- vs Baseline — How much better (+) or similar (~) compared to simply repeating yesterday's value.
- Confidence Range — % of actual values that fell within our predicted high/low range. Higher = better.
- Grade — Overall score: A = excellent, B = good, C = needs improvement.